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Quest 6: The Nullspiracy

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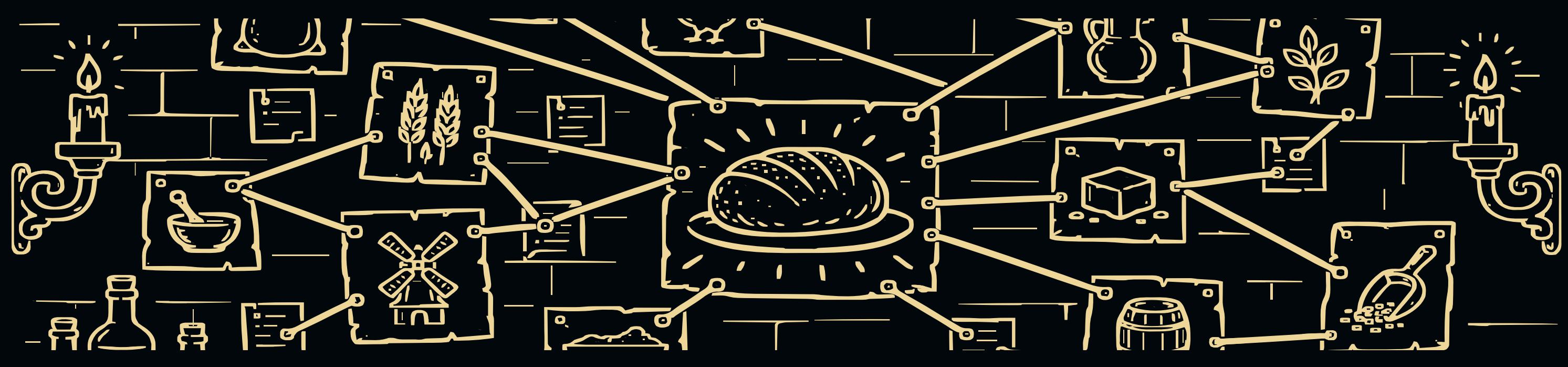

Part I

Story section

Understanding all the stamps used on the elven maps took us two full days. By the time Legolink finally declared that we were ready to continue, we were tired, hungry and no longer certain that getting lost would have been any more complicated. We stopped at a small roadside inn called The Tipsy Turnip.

Everything seemed perfectly ordinary until the food arrived. Frodoff suddenly became restless. He kept looking around the room, shifting in his chair and glancing nervously at his backpack. A moment later, a thin trail of black smoke began to rise from the egg. That was rarely a sign of anything pleasant.

We carefully moved the egg closer to each dish to see which one caused the strongest reaction. When we brought it near the bread, the smoke immediately grew thicker and darker. Legolink took a closer look at the innkeeper and the other guests. Their tired faces, sudden bursts of anger and suspicious glances no longer seemed quite so ordinary. The contaminated bread had probably been served throughout the area for some time.

Luckily, the kingdom required the origin of every ingredient to be recorded. The innkeeper therefore had documents showing which ingredients had been used to make each product, which ingredients had been used to make those ingredients, and so on.

The innkeeper brought us a large collection of documents describing the origin of every product and ingredient used to prepare our food. To begin with, we examined only a small part of those records (your notes). Each product and ingredient had been assigned a four-letter identifier, presumably because actual names would have made the records far too easy to understand.

Each record had the following format:
`ITEM:INGREDIENT_1,INGREDIENT_2,...`

For example:
`BRTS:FLWR,SLTT,MLKK`

This record meant that the item identified as `BRTS` **had been made directly from the ingredients identified as** `FLWR`, `SLTT` and `MLKK`.

Those ingredients could, in turn, have been made from other ingredients. The same ingredient could be used in several different items, and some chains of ingredients could even lead back to an item that we had already examined. If an identifier never appeared on the left-hand side of a record, it represented a basic ingredient that had not been made from anything else.

Frodoff suddenly pointed at one particularly suspicious identifier: `NULL`. He wondered whether it might somehow be connected to Nulldor. That was enough reason for us to investigate it first. **Starting with `NULL`, we searched for every ingredient used to make it,** both directly and indirectly. We counted each different ingredient only once, even if it was used in several items or could be reached through several different chains. We did not count `NULL` itself, even if a chain of ingredients led back to it.

Example based on the following notes:

```
BRDS:FLWR,SLTT,YSTS
FLWR:SLTT,YSTS
SLTT:YSTS,MLKK,BTTR
NULL:BRDS,FLWR,SLTT
YSTS:BRDS,FLWR
MLKK:SPCS,NULL
HNYN:BRDS,FLWR,SLTT,YSTS
```

The item identified as `NULL` had been made directly from three ingredients: `BRDS`, `FLWR` and `SLTT`.

We then examined the records for those ingredients. This led us to three additional ingredients: `YSTS`, `MLKK` and `BTTR`. The record for `MLKK` led us to one final ingredient, `SPCS`.

The ingredients used directly or indirectly to make `NULL` were: `BRDS`, `FLWR`, `SLTT`, `YSTS`, `MLKK`, `BTTR` and `SPCS`.

There were therefore `7` different ingredients used to make `NULL`.

How many different ingredients were used directly or indirectly to make `NULL`?

Your notes for this part:

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Part I solved with answer: 40

Check your progress

Part II

We carefully examined every ingredient used to make `NULL`, but found nothing suspicious. None appeared to contain even the smallest trace of dark magic.

We decided to investigate the records from the opposite direction. This time, we examined a larger selection of the documents (your notes). Instead of checking what had been used to make `NULL`, we wanted to find out which other products had been made using it. If the contamination had already spread through the local supply chain, this would show us how serious the problem had become.

Example based on the following notes:

```
BRDS:FLWR,SLTT,YSTS
FLWR:SLTT,YSTS
SLTT:YSTS,MLKK,BTTR
NULL:BRDS,FLWR,SLTT
YSTS:BRDS,FLWR
MLKK:SPCS,NULL
HNYN:BRDS,FLWR,SLTT,YSTS
```

The product identified as `MLKK` used `NULL` directly.

The product `SLTT` used `MLKK`, so it used `NULL` indirectly. Following the records further showed that `BRDS`, `FLWR`, `YSTS` and `HNYN` also used `NULL` indirectly.

The products that used `NULL` directly or indirectly were: `MLKK`, `SLTT`, `BRDS`, `FLWR`, `YSTS` and `HNYN`.

There were therefore `6` different products that used `NULL`.

How many different products used `NULL` directly or indirectly?

Your notes for this part:

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Part II solved with answer: 814

Check your progress

Part III

We still found nothing suspicious. The only unusual thing was that `NULL` seemed to be almost everywhere. However, the egg did not react to every product containing it, so `NULL` alone could not have been the cause of the contamination.

Legolink asked why we were treating every occurrence of `NULL` as if it represented the same ingredient. To him, the differences were obvious. Some identifiers used the Old Elvish form of the letter `L` instead of `L`, while others used the Old Elvish form of the letter `U` instead of `U`.

None of us could see any difference. Legolink sighed and labelled each occurrence for us as one of three different versions: `NULLv1`, `NULLv2` and `NULLv3`.

We immediately noticed that the suspicious bread contained two different versions of `NULL`. Perhaps each version was harmless on its own, but combining different versions caused the dangerous effect.

Using all the documents we had received (your notes), we decided to **find every product that used at least two different versions of `NULL`**, either directly or indirectly.

Example based on the following notes:

```
MEAL:BRDS,CAKE,DOUG
BRDS:DOUG,FLWR
CAKE:DOUG,FLWR,NULLv2,NULLv3
DOUG:FLWR,YSTS,SLTT,NULLv1
FLWR:BRDS,DOUG
YSTS:SPCS
SOUP:BRDS,CAKE,DOUG,FLWR
```

The product `DOUG` used `NULLv1` directly. Following the records further showed that `BRDS`, `CAKE`, `FLWR`, `MEAL` and `SOUP` also used `NULLv1` indirectly.

The product `CAKE` used both `NULLv2` and `NULLv3` directly. The products `MEAL` and `SOUP` used both of these versions indirectly through `CAKE`.

In addition, `CAKE` used `NULLv1` indirectly through `DOUG`. This meant that `CAKE`, `MEAL` and `SOUP` each used at least two different versions of `NULL`.

The products `DOUG`, `BRDS` and `FLWR` used only `NULLv1`, so they were safe.

There were therefore `3` different products that used at least two different versions of `NULL`.

How many different products used at least two different versions of `NULL`, directly or indirectly?

Your notes for this part:

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Part III solved with answer: 1622

Check your progress

Puzzle solved! Don’t stop now!

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